

PH-102

After Mid Semester Exams

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Chapter-6

Magnetic Fields in Matter

- **Magnetization**
- **Torques and Forces on Magnetic Dipoles**
- **Bound currents (Volume and Surface)**
- **Effects of Magnetic field on Atomic Orbits**

Magnetic Torque

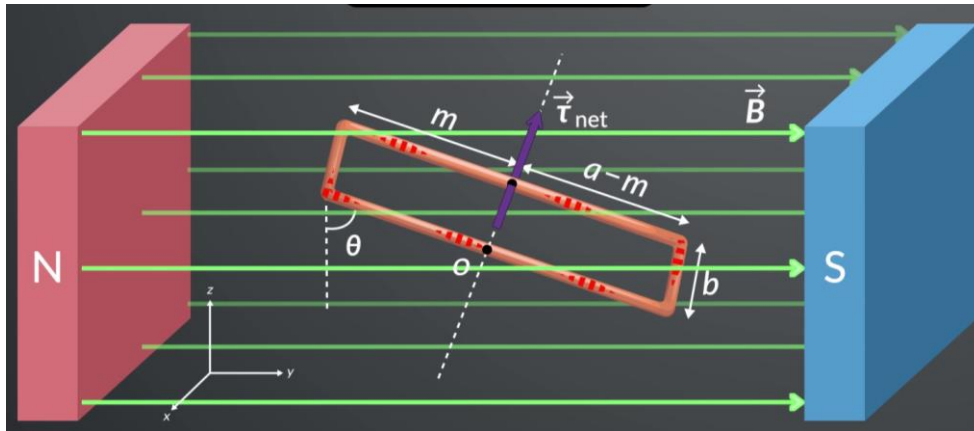
There exist many types of materials which, when placed in an external magnetic field $\vec{B}(\mathbf{r})$ become magnetized.

At the microscopic level, internal atomic/molecular magnetic dipole moments $\vec{m}(\text{atom})$ or $\vec{m}(\text{molecular})$, which, in the presence of the external magnetic field $\vec{B}_{ext}(\mathbf{r})$ produce magnetic torques given by the relation:

Analogous to

$$\vec{\tau}(\mathbf{r}) = \vec{m}(\mathbf{r}) \times \vec{B}_{ext}(\mathbf{r})$$

$$\vec{\tau}(\mathbf{r}) = \vec{p}(\mathbf{r}) \times \vec{E}_{ext}(\mathbf{r})$$



Torque on a Current Loop

$$\vec{\tau}_{net} = -IabB \sin \theta \hat{i}$$

Analogous to

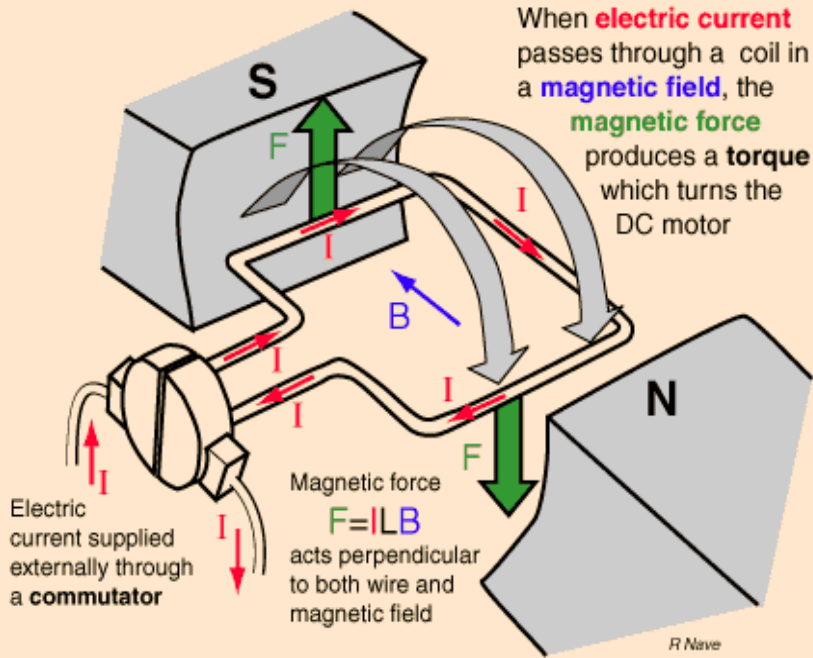
$$\vec{\tau}(\mathbf{r}) = \vec{m}(\mathbf{r}) \times \vec{B}_{ext}(\mathbf{r})$$

$$\vec{\tau}(\mathbf{r}) = \vec{p}(\mathbf{r}) \times \vec{E}_{ext}(\mathbf{r})$$

This is analogous to the situation associated with dielectric materials where electrostatic torques $\vec{\tau}(\mathbf{r}) = \vec{p}(\mathbf{r}) \times \vec{E}_{ext}(\mathbf{r})$ act on individual atomic/molecular electric dipole moments $\vec{p}(\text{atom})/\vec{p}(\text{molecular})$ in an external electric field $\vec{E}_{ext}(\mathbf{r})$ resulting in a net, macroscopic electric polarization, $\vec{P}(\mathbf{r})$.

DC-Motor Operation

This is an active graphic. Click on bold type for further illustration.



Torque on a Current Loop

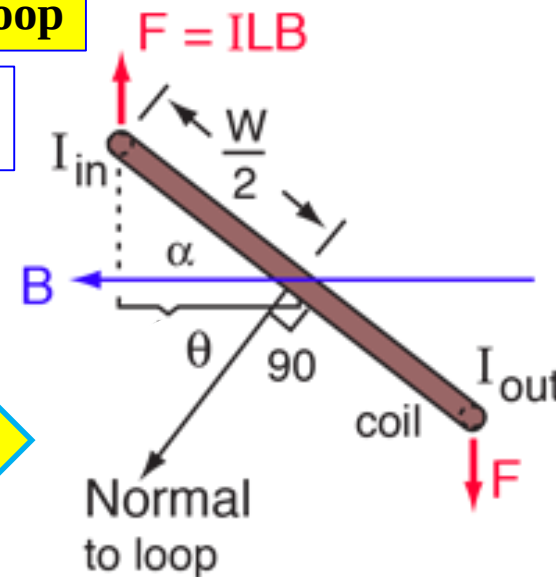
$$\tau = BILW \sin \theta$$

$$\tau = \mu B \sin \theta$$

Lever arm =

$$\frac{W}{2} \cos \alpha$$

$$= \frac{W}{2} \sin \theta$$



$$W = - \int_0^\pi \tau d\theta = - \int_0^\pi \mu B \sin \theta d\theta = -\mu B \cos \theta \Big|_0^\pi = 2\mu B$$

Magnetization, $\vec{M}(r)$

The torque acting on the individual Atomic/Molecular dipole moments, causes a net alignment of the atomic/molecular magnetic dipole moments $\vec{m}(\text{atom}) / \vec{m}(\text{molecular})$, which in turn results in a net macroscopic magnetic polarization, also known as the magnetization, $\vec{M}(r)$.

$$M = m/V$$

Magnetic Dipole moment

Volume of the material

SI Units of M = Amperes per Meter (cgs emu.g⁻¹)

$K(r)$ (Amp/meter)

Magnetic Susceptibility χ_m



$$\chi_m = M/H$$

Dimensionless quantity

H is the magnetic field strength, also measured in Amp/m.

Differential Susceptibility

$$d\chi_m = dM/dH$$

$$B = \mu_o(H+M) = \mu_o(1+\chi_m H) = \mu M$$

$$B = \mu_o(H+M) = \mu_o(1+\chi_m)H = \mu H$$

$$B = \mu\mu_r M$$

Magnetic permeability

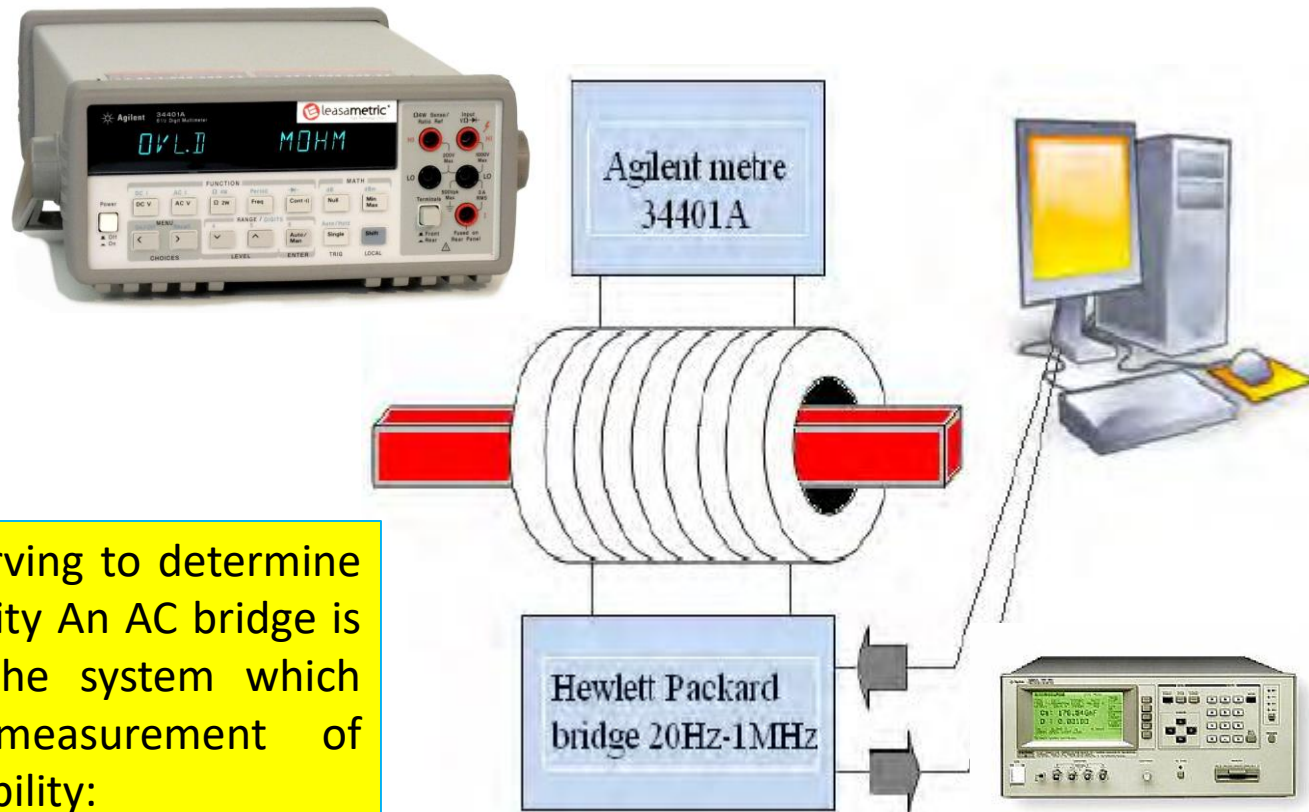
Relative permeability of the material

$$\mu = \mu_o(1+\chi_m)$$

$$\frac{\mu}{\mu_o} - 1 = \chi_m$$

$$\mu_r - 1 = \chi_m$$

Measurement apparatus serving to determine relative magnetic permeability An AC bridge is the chief component of the system which provides for accurate measurement of combined magnetic permeability:

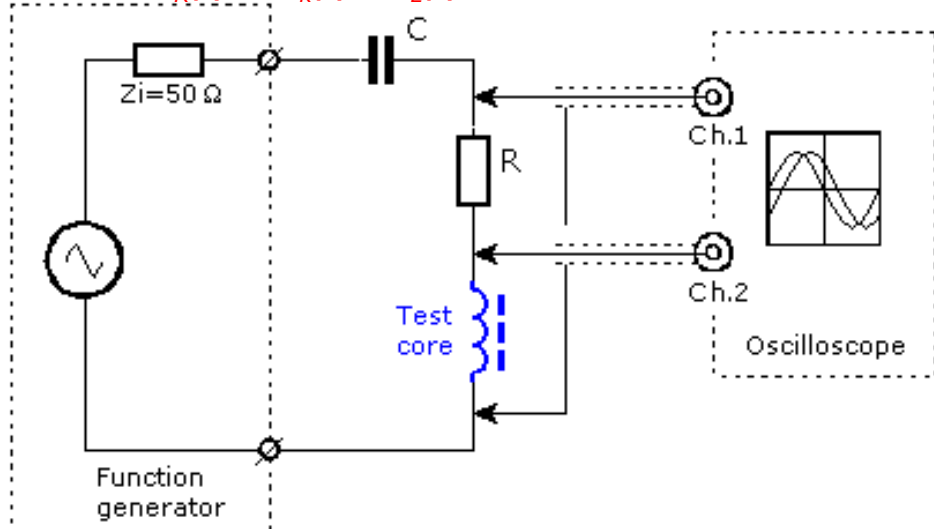


The measurement of magnetic permeability

To measure the permeability the coil must be provided with one coil. The coil must have approximately one turn for every millimeter magnetic path length. The number of turns N must be counted carefully.

Clean the surfaces of dividable cores very carefully. All the smudge and fluff has to be removed. And the core halves must be clamped straight to each other with a sufficient force.

Set the function generator to output a sinewave with a frequency f between 100 and 1000 Hz. Adjust the output voltage to a large as possible amplitude where there isn't any distortion is visible. Optionally adjust the frequency so that both the voltage and current can be read with a high as possible accuracy. The oscilloscope shows now two signals: the voltage across the inductor V_L and the terminal voltage of the function generator V_k , namely the voltage across the resistor and inductor. To obtain the voltage across the resistor V_R , these two signals must be subtracted by the oscilloscope, and will results in a third waveform that corresponds to the voltage across the resistor R : $v_R(t) = v_k(t) - v_L(t)$. With the effective resistor voltage the inductor current I_L can be determined:



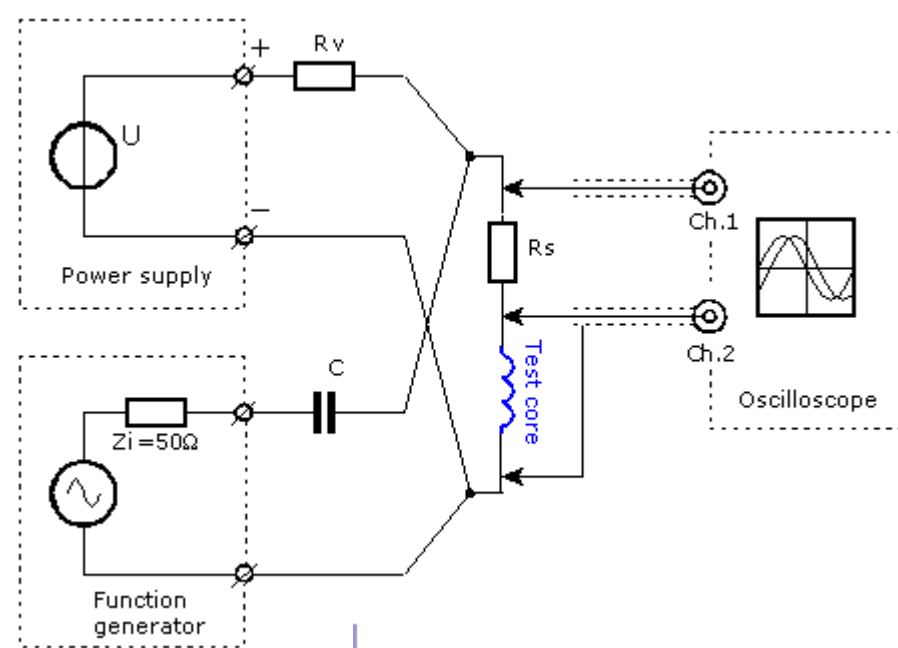
$$I_L := \frac{V_R}{R}$$

$$\mu_c := \frac{V_L \cdot l_c}{I_L \cdot A_c \cdot N^2 \cdot 2 \cdot \pi \cdot f}$$

The last step is to calculate the relative permeability: $\mu_r = \mu_c / \mu_0$. Where the permeability of vacuum $\mu_0 = 4 \cdot \pi \cdot 10^{-7}$

Driving dependence permeability

Magnetic materials used for cores in inductors and transformers are non-linear. The permeability is strongly dependable of the field strength. This dependency can be measured and plotted in a graph.



Magnetic core length $l_c := 92.2 \cdot 10^{-3} \text{ m}$

Number of turns ($N_p = N_s$) $N := 49$

Permeability vacuum $\mu_0 := 4 \cdot \pi \cdot 10^{-7} \frac{\text{H}}{\text{m}}$

Measure frequency $f := 1000 \text{ Hz}$

The measurement values:

Direct current I_{dc} , Alternating voltage V_{ac} , Alternating current I_{ac}

$I_{dc} :=$	1	$V_{ac} :=$	405.3	$I_{ac} :=$	8.40
	2		406.4		8.49
	5		408.0		8.39
	10		409.0		8.26
	20		406.7		8.23
	50		401.8		8.61
	100		378.9		10.36
	200		247.6		15.65
	500		68.44		17.86
	1000		21.69		17.81

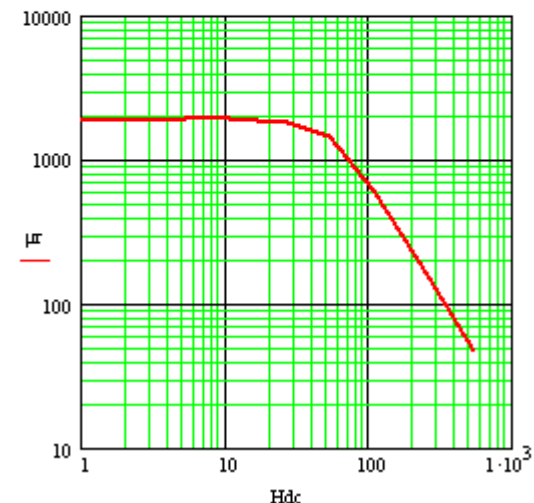
Processing measurement data

First the core parameters must be defined: cross-sectional area A_c , magnetic path length l_c , the number of turns N , and also the frequency f .

For every measure point the DC-current I_{dc} must be set and the AC-voltage V_{ac} and DC-current I_{ac} must be measured and listed. This procedure has to be repeated until the required field strength range is covered. From every measurement the field strength H and the relative permeability μ_r can be calculated. The series of calculated values can be plotted onto a log-log-graph.

Field strength $H_{dc} := \frac{I_{dc} \cdot N}{l_c}$

Relative permeability $\mu_r := \frac{V_{ac} \cdot l_c}{I_{ac} \cdot A_c \cdot N^2 \cdot 2 \cdot \pi \cdot f \cdot \mu_0}$

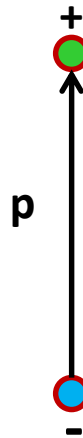


Earlier Magnetostatics Model based on Magnetic charges

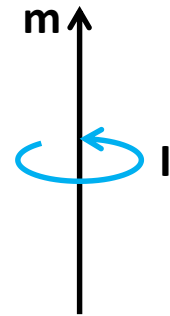
□ **Gilbert Model:** Earlier physicists who thought magnetic dipoles consisted of positive and negative magnetic "charges" (north and south "poles," they called them), separated by a small distance, just like electric dipoles.



Magnetic dipole
(Gilbert model)



Electric dipole



Magnetic dipole
(Ampère model)

The Field of a Magnetized Object

Bound Currents

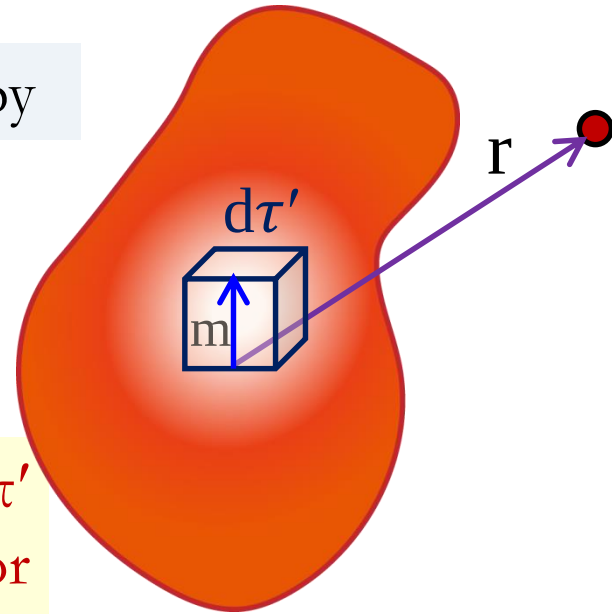
Suppose we have a piece of magnetized material; the magnetic dipole moment per unit volume, $\mathbf{M}$, is given. What field does this object produce?

The vector potential of a single dipole $\mathbf{m}$ is given by

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

In the magnetized object, each volume element $d\tau'$ carries a dipole moment $\mathbf{M} d\tau'$, so the total vector potential is

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \frac{\overrightarrow{M(r')} \times \hat{r}}{r^2} d\tau'$$

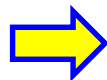


Equation 1

This integral can be cast in a more illuminating form by exploiting the identity

$$\nabla' \frac{1}{r} = \frac{\hat{r}}{r^2}$$

$$\vec{A}(r) = \frac{\mu_0}{4\pi} \int \frac{\vec{M}(r') \times \hat{r}}{r^2} d\tau'$$



$$\vec{A}(r) = \frac{\mu_0}{4\pi} \int \left[\vec{M}(r') \times \left(\nabla' \frac{1}{r} \right) \right] d\tau'$$

Using the relation

$$\nabla \times (f \vec{A}) = f(\nabla \times \vec{A}) - \vec{A} \times (\nabla f)$$

Here

$$f = \frac{1}{r}$$

$$\vec{A} = \vec{M}(r')$$

$$\nabla' \times \left(\frac{1}{r} \vec{M}(r') \right) = \frac{1}{r} (\nabla' \times \vec{M}(r')) - \vec{M}(r') \times (\nabla' \frac{1}{r})$$

$$\nabla' \times \left(\frac{1}{r} \overrightarrow{M(r')} \right) = \frac{1}{r} \left(\nabla' \times \overrightarrow{M(r')} \right) - \underbrace{\overrightarrow{M(r')} \times \left(\nabla' \frac{1}{r} \right)}$$

$$\overrightarrow{M(r')} \times \left(\nabla' \frac{1}{r} \right) = \frac{1}{r} \left(\nabla' \times \overrightarrow{M(r')} \right) - \nabla' \times \left(\frac{1}{r} \overrightarrow{M(r')} \right)$$

Substituting this value in
the main integral

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \left[\overrightarrow{M(r')} \times \left(\nabla' \frac{1}{r} \right) \right] d\tau'$$

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \left[\frac{1}{r} \left(\nabla' \times \overrightarrow{M(r')} \right) - \nabla' \times \left(\frac{1}{r} \overrightarrow{M(r')} \right) \right] d\tau'$$

Using the relation

$$\int_V (\nabla \times \vec{v}) d\tau = - \oint_S \vec{v} \times d\vec{a}$$

The second term in this integration will become:

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \left\{ \int \left[\frac{1}{r} \left(\nabla' \times \overrightarrow{M(r')} \right) \right] d\tau' + \oint_s \frac{1}{r} \left[\overrightarrow{M(r')} \times d\mathbf{a}' \right] \right\}$$

The first term looks just like the potential of a volume current

$$\overrightarrow{J}_b = \nabla \times \overrightarrow{M}$$

While the second term looks like the potential of a surface current

$$\overrightarrow{K}_b = \overrightarrow{M} \times \hat{n}$$

Where $\hat{n}$ is the normal unit vector. With these definitions

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int_s \left[\frac{J_b(r')}{r} \right] d\tau' + \frac{\mu_0}{4\pi} \oint_s \frac{K_b(r')}{r} da'$$

$$\vec{A}(r) = \frac{\mu_0}{4\pi} \int_S \left[\frac{J_b(r')}{r} \right] d\tau' + \frac{\mu_0}{4\pi} \oint_S \frac{K_b(r')}{r} da'$$

Equation 2

From this relation we can see that the potential of a magnetized object is the same as would be produced by a volume current $\vec{J}_b = \nabla \times \vec{M}$ throughout the material plus a surface current $\vec{K}_b = \vec{M} \times \hat{n}$, on the boundary.

Instead of integrating the contributions of all the infinitesimal dipoles, as in Eq. 1, we first determine these bound currents, and then find the field they produce, in the same way we would calculate the field of any other volume and surface currents.

Notice the striking parallel with the electrical case: there the field of a polarized object was the same as that of a bound volume charge $\rho_b = -\nabla \cdot \vec{P}$ plus a bound surface charge $\sigma_b = \vec{P} \cdot \hat{n}$.

Physical Interpretation of Bound Currents

This section discuss how these bound currents arise physically.

Figure 1 depicts a thin slab of uniformly magnetized material, with the dipoles represented by tiny current loops.

Notice that all the "internal" currents cancel: every time there is one going to the right, a contiguous one is going to the left.

However, at the edge there is no adjacent loop to do the canceling. The whole thing, then, is equivalent to a single ribbon of current I flowing around the boundary (Figure 2)

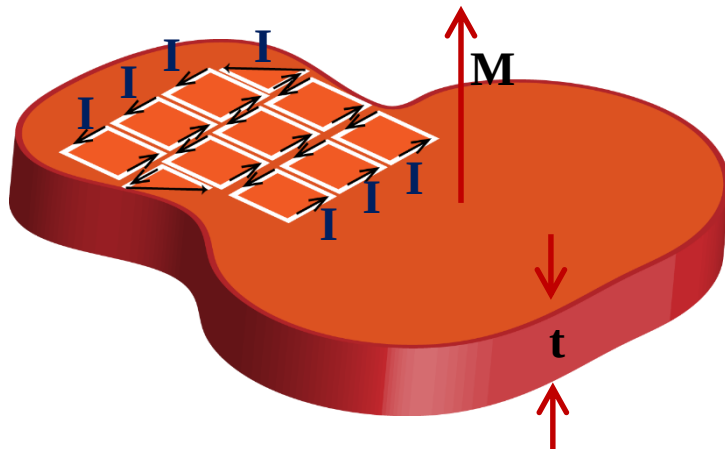


Figure 1

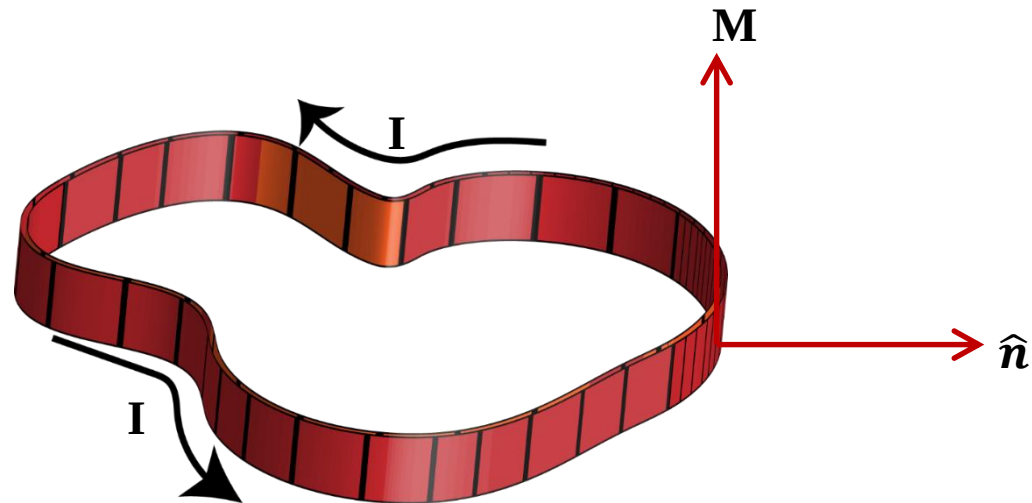


Figure 2

Uniform Magnetization $\mathbf{M}$

What is this current, in terms of $\mathbf{M}$?

Say that each of the tiny loops has area a and thickness t (Figure 3)

In terms of the magnetization $\mathbf{M}$, its dipole moment is

$$\mathbf{m} = \mathbf{M} \times a \times t$$

In terms of the circulating current I ,

$$m = Ia$$

Therefore

$$I = Mt$$

So the surface current is

$$K_b = \frac{I}{t}$$

$$K_b = \frac{I}{t} = M \longrightarrow \vec{K}_b = \vec{M} \times \hat{n}$$

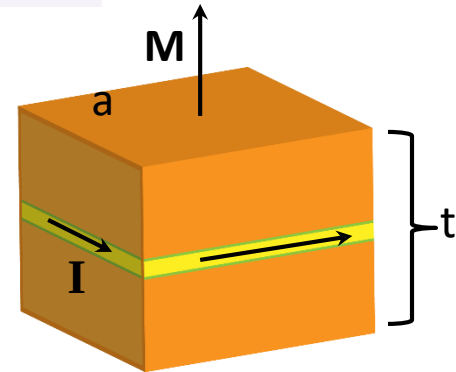


Figure 3

Using the outward-drawn unit vector (Figure 2), the direction of $\vec{K}_b$ is conveniently indicated by the cross product:

$$\vec{K}_b = \vec{M} \times \hat{n}$$

This expression also records the fact that there is no current on the top or bottom surface of the slab; here $\vec{M}$ is parallel to $\hat{n}$, so the cross product vanishes.

This bound surface current is exactly what we obtained in previous

It is a peculiar kind of current, in the sense that no single charge makes the whole trip on the contrary, each charge moves only in a tiny little loop within a single atom.

Nevertheless, the net effect is a macroscopic current flowing over the surface of the magnetized object. We call it a "bound" current to remind ourselves that every charge is attached to a particular atom, but it's a perfectly genuine current, and it produces a magnetic field in the same way any other current does.

Non-uniform Magnetization

When the magnetization is non-uniform, the internal currents no longer cancel.

Figure 4 shows two adjacent chunks of magnetized material, with a larger arrow on the one to the right suggesting greater magnetization at that point. On the surface where they join there is a net current in the x-direction, given by

$$I_x = [M_Z(y + dy) - M_Z(y)]dz$$

$$I_x = \left[M_Z(y) + \frac{\partial M_Z}{\partial y} dy - M_Z(y) \right] dz$$

$$I_x = \frac{\partial M_Z}{\partial y} dy dz$$

$\vec{K}_b = \vec{M} \times \hat{n}$

M_Z at the location of y

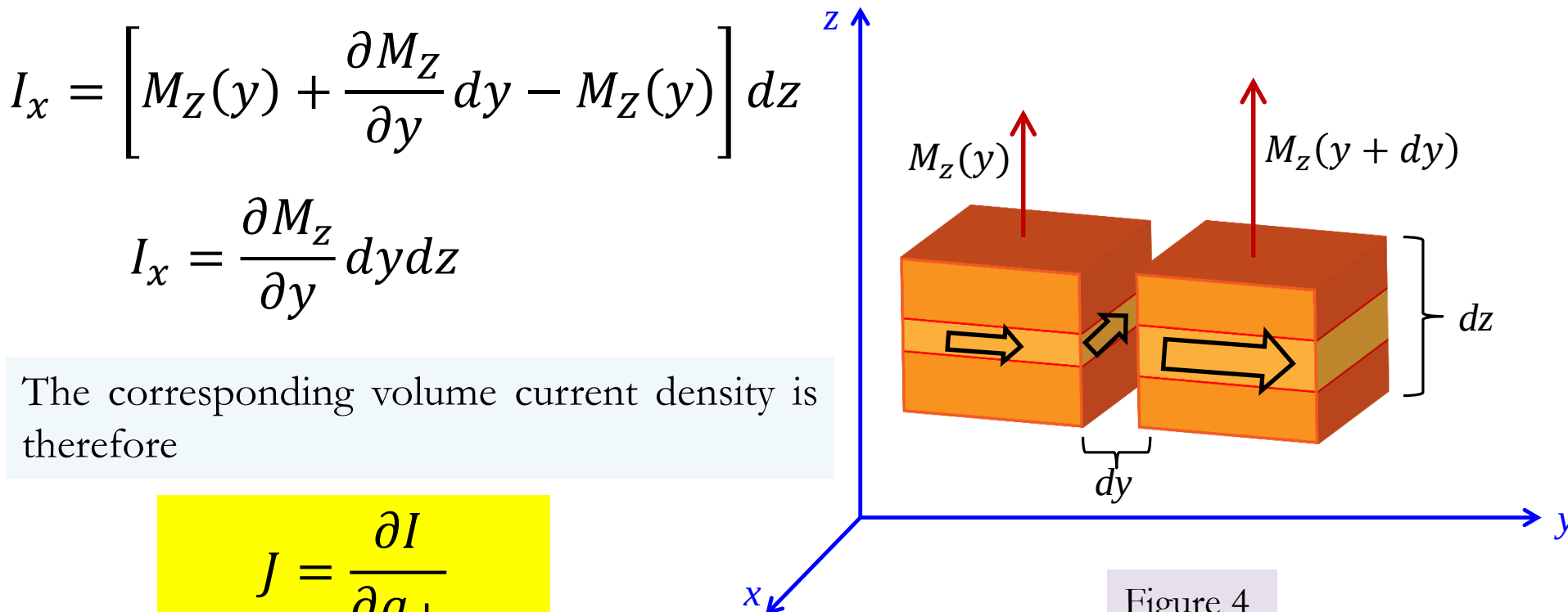
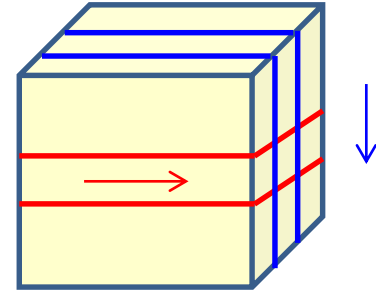


Figure 4

The corresponding volume current density is therefore

$$J = \frac{\partial I}{\partial a_{\perp}}$$

$$(J_b)_x = \frac{\partial M_z}{\partial y}$$



Using the same argument, a non-uniform magnetization in the y-direction would contribute an amount $-\frac{\partial M_y}{\partial z}$ (along $-z$ direction)

So, the total contribution to $(J_b)_x$

$$(J_b)_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z}$$

In general

$$\vec{J}_b = \nabla \times \vec{M}$$

Also, like any other steady current, J_b should obey the conservation law

$$\nabla \cdot \vec{J}_b = 0$$

$$\nabla \cdot (\nabla \times \vec{M}) = 0$$

The divergence of a curl is always zero.

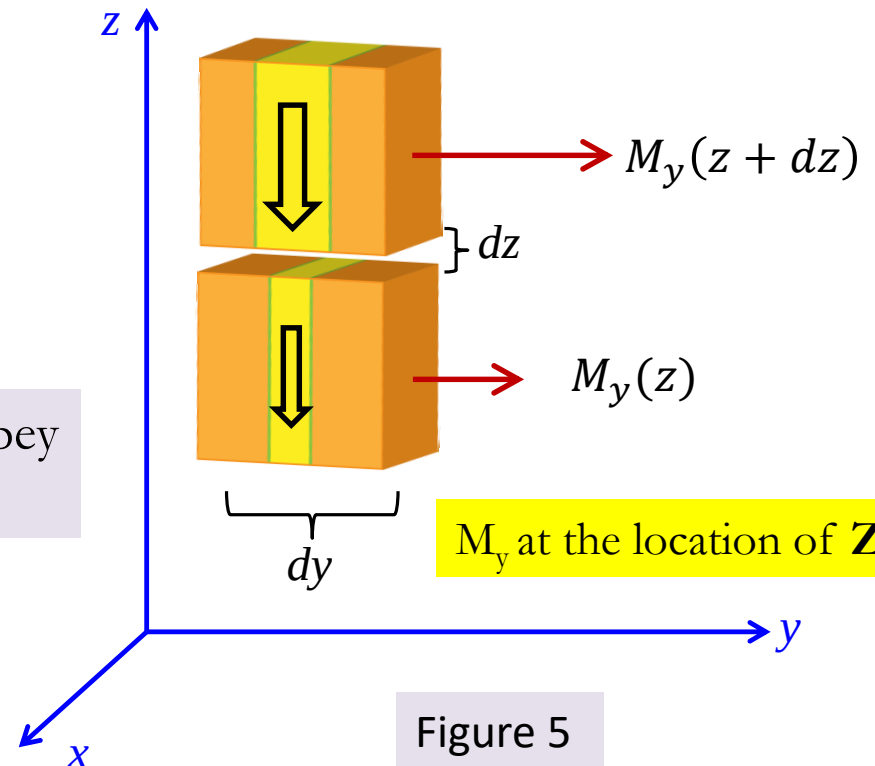


Figure 5

The Auxiliary Field H

Ampere's law in Magnetized Materials

The effect of magnetization is to establish bound currents

$$\vec{J}_b = \nabla \times \vec{M}$$

Volume bound current

$$\vec{K}_b = \vec{M} \times \hat{n}$$

Surface bound current

The field due to magnetization of the medium = The field produced by these bound currents.

We are now ready to put everything together: the field attributable to bound currents, plus the field due to everything else-which I shall call the free current.

The free current might flow through wires imbedded in the magnetized substance or, if the matter is a conductor, through the material itself.

The total current can be written as

$$\vec{J} = \vec{J}_b + \vec{J}_f$$

Bound Current



Free Current

Free Current

The free current is there because somebody hooked up a wire to a battery since it involves actual transport of charge.

Bound Current

The bound current is there because of magnetization, it results from the conspiracy of many aligned atomic dipoles.

Now the Ampere's law can be written as

$$\frac{1}{\mu_0} (\nabla \times \vec{B}) = \mathbf{J} = \mathbf{J}_b + \mathbf{J}_f$$

$$\vec{J}_b = \nabla \times \vec{M}$$

$$\frac{1}{\mu_0} (\nabla \times \vec{B}) = \nabla \times \vec{M} + \mathbf{J}_f$$

Collecting together the two curls

$$\left(\nabla \times \left(\frac{1}{\mu_0} \vec{B} - \vec{M} \right) \right) = J_f$$

The quantity in parentheses is designated by the letter **H**

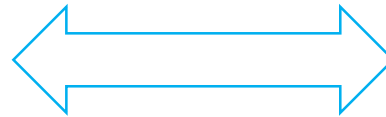
Differential Form

$$\nabla \times \vec{H} = J_f$$

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$$

Integral Form

$$\oint \vec{H} \cdot d\vec{l} = I_{f_{enclosed}}$$



H plays a role in magneto-statics analogous to **D** in electrostatics. Just as **D** allowed us to write Gauss's law in terms of the free charge alone, **H** permits us to express Ampere's law in terms of the free current alone and free current is what we control directly.

Bound current, like bound charge, comes along for the ride the material gets magnetized, and this results in bound currents; we cannot turn them on or off independently, as we can free currents.

Few important points about H:

- ❑ As it turns out, H is a more useful quantity than D . In the laboratory you will frequently hear people talking about H (more often even than B), but you will never hear anyone speak of D (only E).
- ❑ The reason is this: To build an electromagnet you run a certain (free) current through a coil. The current is the thing you read on the dial, and this determines H (or at any rate, the line integral of H); B depends on the specific materials you used and even, if iron is present, on the history of your magnet.
- ❑ On the other hand, if you want to set up an electric field, you do not plaster a known free charge on the plates of a parallel plate capacitor; rather, you connect them to a battery of known voltage.
- ❑ It's the potential difference you read on your dial, and that determines E (or at any rate, the line integral of E); D depends on the details of the dielectric you're using.
- ❑ If it were easy to measure charge, and hard to measure potential, then you'd find experimentalists talking about D instead of E . So the relative familiarity of H , as contrasted with D , derives from purely practical considerations; theoretically, they're all on equal footing.

A Deceptive (Misleading) Parallel

Curl alone does not determine a vector field, you must know the divergence as well.

❑ For magnetic field $\mathbf{B}$

$$\nabla \cdot \mathbf{B} = 0$$

❑ Does it true for $\mathbf{H}$ also ?

The divergence of $\mathbf{H}$ is not, in general, zero.

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$$

$$\nabla \cdot \vec{H} = \frac{1}{\mu_0} \nabla \cdot \vec{B} - \nabla \cdot \vec{M}$$

Taking the divergence

Since

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \cdot \vec{H} = -\nabla \cdot \vec{M}$$

Only when the divergence of $\mathbf{M}$ vanishes is the parallel between $\mathbf{B}$ and $\mu_0 \mathbf{H}$ faithful.

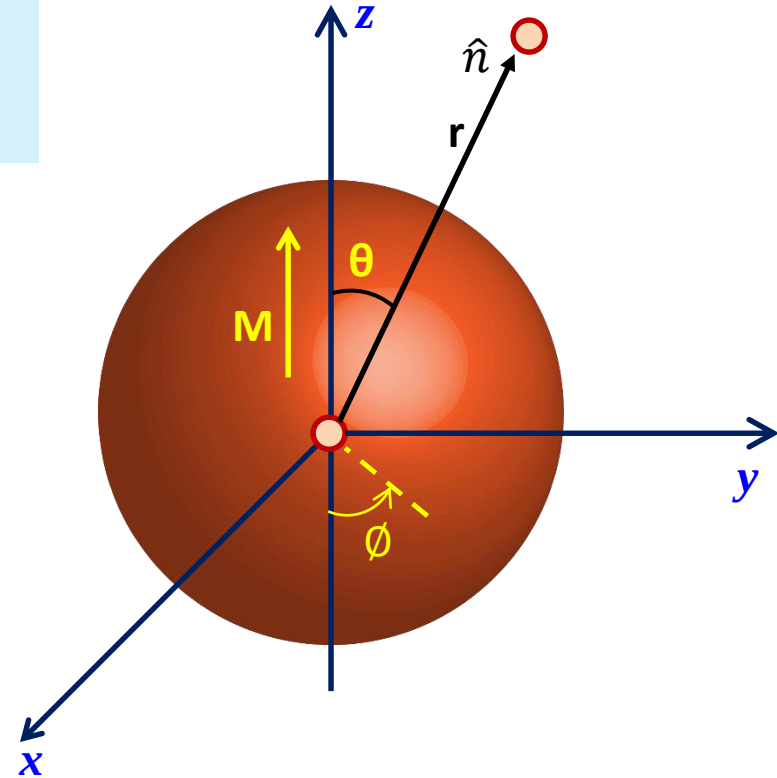
Example: Magnetic field of a uniformly magnetized sphere.

Solution: Choosing the z-axis along the direction of $\vec{M}$, we have

$$\vec{J}_b = \nabla \times \vec{M} = 0$$

$$\vec{K}_b = \vec{M} \times \hat{n}$$

$$K_b = M \sin \theta \hat{\phi}$$



Now, a rotating spherical shell, of uniform surface charge σ , corresponds to a surface current density

$$\vec{K}_b = \sigma \vec{v}$$

$$\mathbf{K}_b = \sigma \vec{V}$$

$$K_b = \sigma \omega R \sin \theta \hat{\phi}$$

The field of a uniformly magnetized sphere is identical to the field of a spinning spherical shell, with the identification of $\sigma \omega R \rightarrow \mathbf{M}$

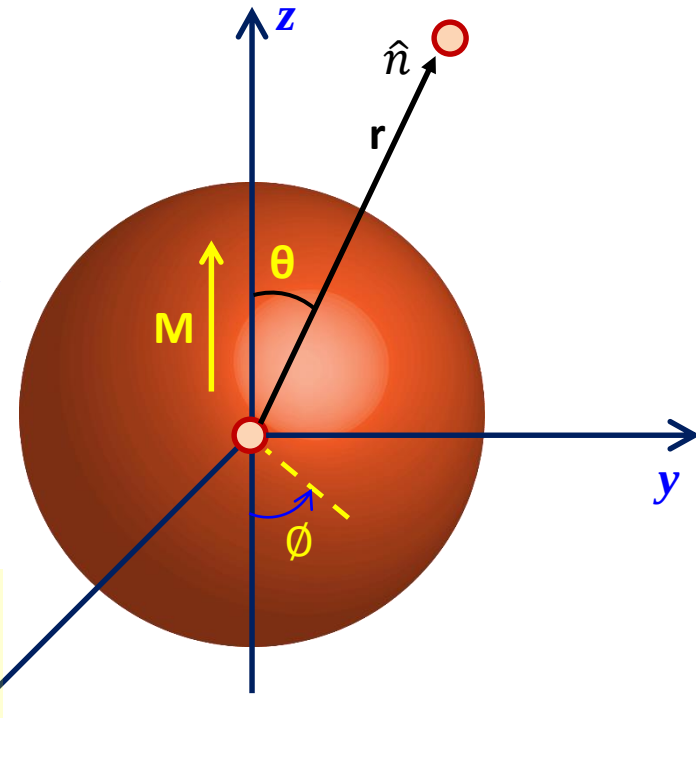
$$K_b = M \sin \theta \hat{\phi}$$

$$\vec{B} = \frac{2\mu_0 \sigma \omega R}{3} \hat{z}$$

Ref: Page 237 & 169
See previous Lecture Slide

Previous result for spherical shell Example. 5.11

$$\mathbf{B} = \frac{2}{3} \mu_0 \mathbf{M}$$



Magnetic field inside the sphere

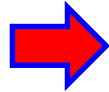
$$B_{\text{inside}} = \frac{2}{3} \mu_0 M$$

$$\mathbf{B}_{\text{outside}} = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(\mathbf{m} \cdot \hat{r})\hat{r} - \mathbf{m}]$$

Inside the sphere, whereas the field outside is the same as that of a pure dipole.

Magnetization 'M' Given (Uniformly magnetized sphere)

Magnetic Dipole Moment



$$\mathbf{m} = \frac{4}{3} \pi R^3 \mathbf{M}$$

Analogy to Electrostatics

The electric field inside a uniformly polarized sphere

$$\mathbf{E} = -\nabla V = -\frac{P}{3\epsilon_0} \hat{\mathbf{z}} = -\frac{1}{3\epsilon_0} \mathbf{P}, \quad \text{for } r < R$$

$$\mathbf{p} = \frac{4}{3} \pi R^3 \mathbf{P}$$

The actual formulas for the two cases are curiously different (2/3 in place of -1/3).

$$\mathbf{A}(\mathbf{r}) = \begin{cases} \frac{\mu_0 R \sigma}{3} (\boldsymbol{\omega} \times \mathbf{r}), \\ \frac{\mu_0 R^4 \sigma}{3 r^3} (\boldsymbol{\omega} \times \mathbf{r}), \end{cases}$$

For points inside the sphere

For points outside the sphere

$$\mathbf{A}(r, \theta, \phi) = \begin{cases} \frac{\mu_0 R \omega \sigma}{3} r \sin \theta \hat{\boldsymbol{\phi}}, & (r \leq R), \\ \frac{\mu_0 R^4 \omega \sigma}{3} \frac{\sin \theta}{r^2} \hat{\boldsymbol{\phi}}, & (r \geq R). \end{cases}$$

$$\mathbf{B} = \nabla \times \mathbf{A} = \frac{2\mu_0 R \omega \sigma}{3} (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}) = \frac{2}{3} \mu_0 \sigma R \omega \hat{\mathbf{z}} = \frac{2}{3} \mu_0 \sigma R \boldsymbol{\omega}.$$

- Take home exercise from D J Griffiths's 3rd edition book

[1] G5.4 [2] G5.5 [3]G5.7 [4]G5.8 [5]G5.9 [6]G5.10 [7]G5.11 [8]G5.12